

Assignment 2 SECPH 2523

QUESTION 1

Table 2.1 STUDENTFINE records data on traffic offenses by students in a college.

1NF=remove repeating grp in UNF(Enter data into each row)

TABLE 2.1 STUDENTFINE

StudentID	Name	LicenseNo	CarReg	ReceiptNo	Date	OffenceCode	Fine(RM)
F1011	CK Fong	P100	JGK9201	00-345	10/03/06	2	30.00
				00-690	15/03/06	1	20.00
				00-810	20/03/06	2	30.00
L782	AdamJames	P290	JJL99	00-459	13/03/06	1	20.00
H762	MalaDewi	P764	JJK7654	00-564	12/03/06	3	40.00
				00-770	19/06/06	2	30.00

- a) Table 2.1 is said to be **unnormalized table**. Why?

It is said to be unnormalized table is because it contains repeating groups. For example, the ReceiptNo column has multiple values for a single student. Other than that, the Date, OffenceCode and Fine are also listed multiple times for the same student

- b) Transform the unnormalized Table 2.1 into a **First Normal Form (1NF) table**. Show the transformed 1 NF table. What is **the primary key** for this new 1 NF table?

PRIMARY KEY are StudentID

StudentID	Name	LicenseNo	CarReg	ReceiptNo	Date	OffenceCode	Fine(RM)
F1011	CK Fong	P110	JGK9201	00-345	10/03/06	2	30.00
F1011	Ck Fong	P110	JGK9201	00-690	15/03/06	1	20.00
F1011	Ck Fong	P110	JGK9201	00-810	20/03/06	2	30.00
L782	Adam James	P290	JJL99	00-564	13/03/06	1	20.00
H762	Mala Dewi	P764	JJK7654	00-564	12/03/06	3	40.00
H762	Mala Dewi	P764	JJK7654	00-770	19/06/06	2	30.00

QUESTION 2

2NF: Eliminate partial dependencies.(split) 3NF: Eliminate transitive dependencies(indirect dependencies)

The following table is un-normalized table that is created to manage a consulting company's operations. Create its normalized forms (1NF, 2NF, and 3NF).

Note: Define the PK and FK for each table:

Project number	Project name	Employee number	Employee name	Rate category	Hourly rate
1023	Madagascar travel site	11	Vincent Radebe	A	\$60
		12	Pauline James	B	\$50
		16	Charles Ramoraz	C	\$40
1056	Online estate agency	11	Vincent Radebe	A	\$60
		17	Monique Williams	B	\$50

1NF:

The Primary Key (PK) is the Project number

Project number	Project name	Employee number	Employee name	Rate category	Hourly rate
1023	Madagascar travel site	11	Vincent Radebe	A	\$60
1023	Madagascar travel site	12	Pauline James	B	\$50
1023	Madagascar travel site	16	Charles Ramoraz	C	\$40
1056	Online estate agency	11	Vincent Radebe	A	\$60
1056	Online estate agency	17	Monique Williams	B	\$50

2NF**PK:** Project number

Project number	Project name
1023	Madagascar travel site
1023	Madagascar travel site
1023	Madagascar travel site
1056	Online estate agency
1056	Online estate agency

PK: Employee number

Employee number	Employee name	Rate category	Hourly rate
11	Vincent Radebe	A	\$60
12	Pauline James	B	\$50
16	Charles Ramoraz	C	\$40
11	Vincent Radebe	A	\$60
17	Monique Williams	B	\$50

3NF

PK:Project Number

Project number	Project name
1023	Madagascar travel site
1023	Madagascar travel site
1023	Madagascar travel site
1056	Online estate agency
1056	Online estate agency

PK: Employee Number

FK: Rate Category

Employee number	Employee name	Rate category
11	Vincent Radebe	A
12	Pauline James	B
16	Charles Ramoraz	C
11	Vincent Radebe	A
17	Monique Williams	B

PK=Rate Category

Rate category	Hourly rate
A	\$60
B	\$50
C	\$40
A	\$60
B	\$50

Project number	Employee number	Rate Category
1023	11	A
1023	12	B
1023	16	C
1056	11	A
1056	17	B